

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0591

Humacao 2601 138 kV
March 17, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0591. The Interconnection Customer proposes a 356.8 MW wind facility interconnecting at the Humacao 2601 138 kV substation in the Virginia area. The request was received on March 02, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 201 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,624,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HUMACAO 2601 – RIV ROAD 138 KV Line	STRATHMORE – HUMACAO 2601 138 KV Line	174.7	188	100.0
HUMACAO 2601 – STRATHMORE 138 KV	RIV RD – HUMACAO 2601 138 KV	173.6	189	100.0
RIV ROAD – FRITCH 138 KV	STRATHMORE – HUMACAO 2601 138 KV L	148.2	208	100.0
STRATHMORE – CHIPOLA 138 KV	HUMACAO 2601 – RIV RD 138 KV line	141.8	214	100.0
BEAR ISLAND – MUSCODA INDUSTRIAL 138 KV	BEAR ISLAND – CHANCELLOR ROW 345 KV line	122.3	318	9.4
DPS OCEAN SPRAY – FOUR RIVERS 138 KV	LEAVITT – BIRCHWOOD 345 KV	111.0	242	6.6
LAKESIDE – FRITCH 138 KV	SKAGGS ISLAND JCT 2 – MITCHELL DAM 345 KV line	101.8	303	24.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RIVER ROAD – HUMACAO 2601 138 KV	Rebuild line with higher-capacity conductor	\$2,744,000	100% (\$2,744,000)
HUMACAO 2601 – STRATHMORE 138 KV	Rebuild line with higher-capacity conductor	\$2,669,000	100% (\$2,669,000)
RIV RD – FRITCH 138 KV	Rebuild line with higher-capacity conductor	\$3,112,000	100% (\$3,112,000)
STRATHMORE – CHIPOLA 138 KV	Rebuild line with higher-capacity conductor	\$2,764,000	100% (\$2,764,000)
MUSCODA INDUSTRIAL – BEAR ISLAND 138 KV L	Rebuild line with higher-capacity conductor	\$6,651,000	13% (\$856,000)
DPS OCEAN SPRAY – FOUR RIVERS 138 KV	Rebuild line with higher-capacity conductor	\$5,699,000	39% (\$2,215,000)
LAKESIDE – FRITCH 138 KV	Reconductor line	\$1,147,000	23% (\$264,000)
	Total allocated to project		\$14,624,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.